TCS & AMD AI Hackathon – Participant Handbook

Welcome to the AMD AI Hackathon! This handbook provides all the essential information to help you successfully participate, build, and submit your AI solutions.

OVERVIEW

This hackathon gives developers the opportunity to explore and build AI workloads using AMD's cloud-based AI stack with access to powerful AMD GPUs—no hardware setup required.

All development happens using **cloud-accessible AMD GPUs**. The focus is on experimenting, benchmarking, and running real workloads in a flexible environment.

STEPS TO FOLLOW

This hackathon gives developers the opportunity to explore and build AI workloads using AMD's cloud-based AI stack.

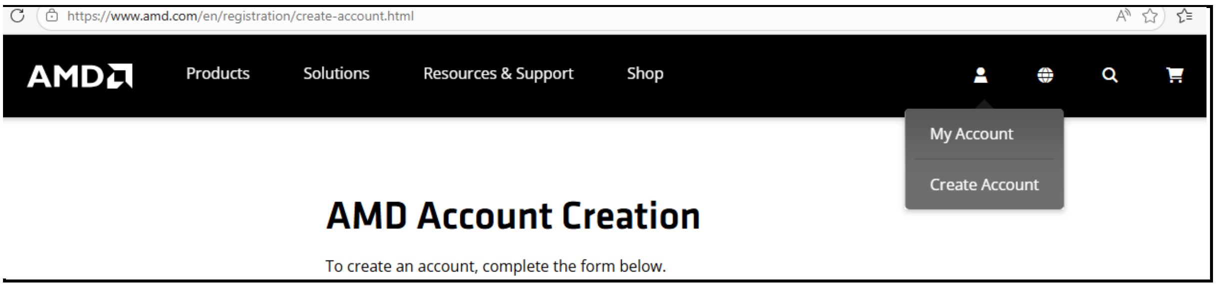
1. **Register to AMD Partner Portal**: https://arena.amd.com/prelogin.html

Register on **arena.amd.com** to access personalized AMD training, track your progress and build expertise in AMD technologies.

2. **Register to AMD Developer Cloud**

The build part of the hackathon will be done on AMD Developer Cloud.

Sign-up for AMD AI Developer Program to participate



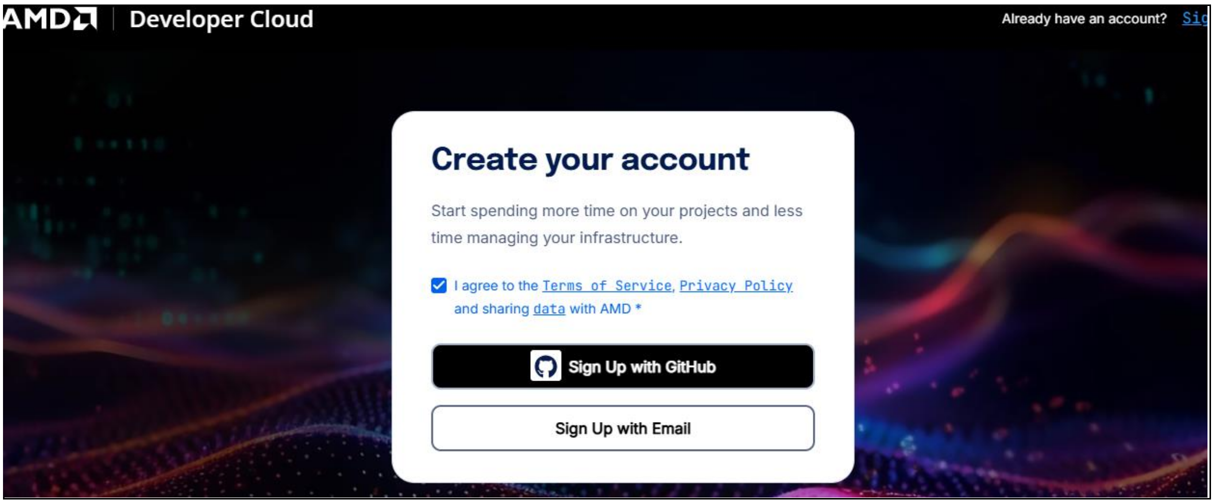
The AMD Developer Cloud offers on-demand access to AMD Instinct GPUs through the cloud. It allows developers to spin up powerful GPU environments in minutes and focus on building and testing AI workloads without infrastructure overhead.

What developers typically use it for:

- Training and fine-tuning machine learning models

- Benchmarking AI workloads on AMD GPUs
- Prototyping AI systems before moving to on-prem infrastructure

Sign-up on AMD Developer Cloud to access AMD GPUs in the cloud



Select the option "Sign Up with Email". Fill your name, email address in the form to proceed.

- $100 in AMD Developer Cloud credits for AMD AI Developer Program members, which will be credited by the 8th of June upon sign-up to AMD AI Developer Program
- How to Get Started on the AMD Developer Cloud and VM creation; Go through this link and familiarize yourself

  https://www.amd.com/en/developer/resources/technical-articles/2025/how-to-get-started-on-the-amd-developer-cloud-.html

- Access the VM from your laptop/desktop using the Web Console (AMD Developer Cloud) and multiple sessions if needed

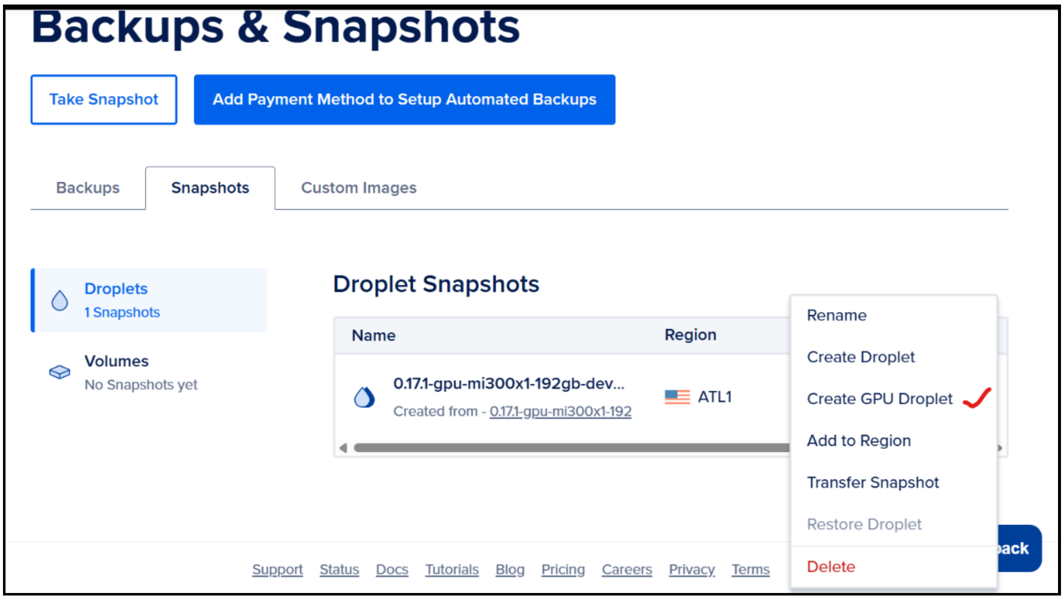
All you need to know about GPU Credits

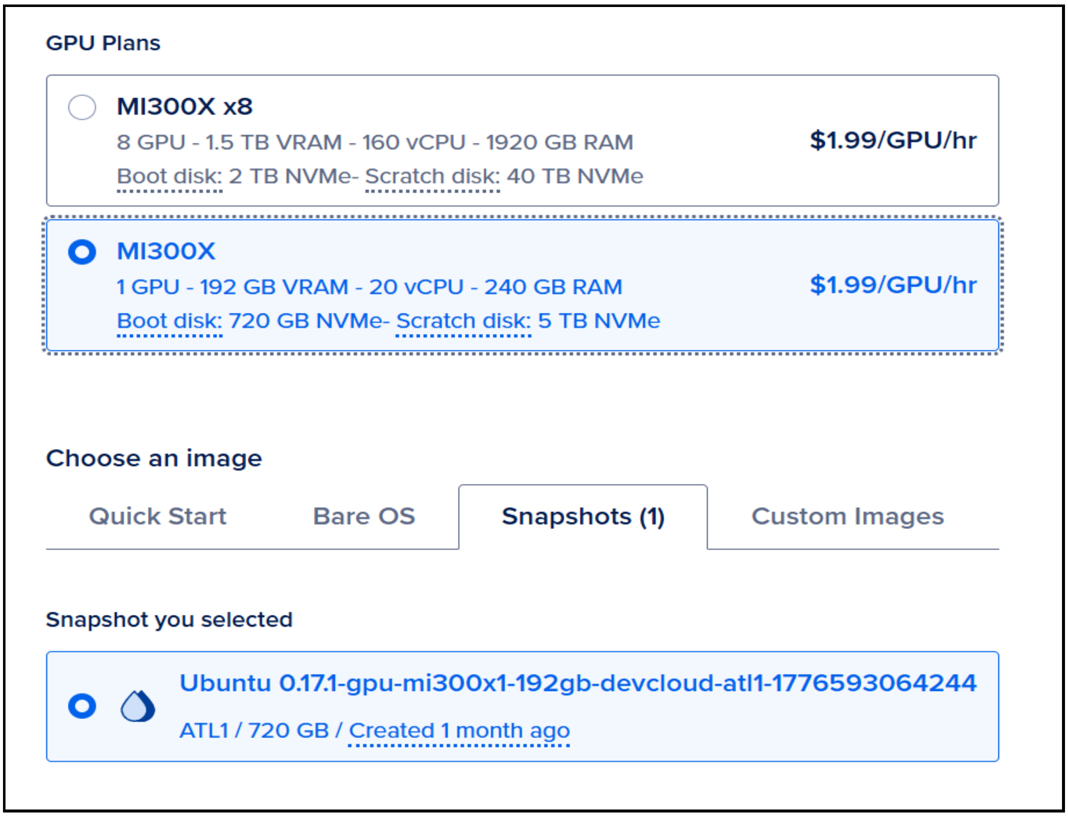
AMD provides participants with cloud-based GPU credits to enable seamless development and experimentation on its high-performance AI infrastructure. These credits can be used to access the AMD Developer Cloud, allowing teams to train, fine-tune and run AI/ML workloads without needing physical hardware.

Participants are encouraged to utilize these credits efficiently by managing resources and stopping unused instances. This ensures maximum productivity while exploring and building scalable AI solutions.

How to utilise the Credits efficiently:

- To ensure that one doesn't run out of GPU credit, remember to snapshot your Droplet / GPU Instance and destroy the virtual machine to avoid consuming credits when not actually doing work. Resume your work from where you left off using the snapshot as the image to launch your droplet.





NOTE: GPU resources may be subject to peak usage constraints during the hackathon. In such cases, temporary unavailability may occur, and participants are advised to retry after some time.

Community/Expert Connect

Stay connected with AMD experts for the duration of the Hackathon to get answers for your technical queries. Install Discord application on your **personal device**.

- Sign-up for AMD Discord, Register using your personal email id & verify it.
- Click on + to Join a Server
- **https://discord.com/invite/CxfXGhdp3J** - Please join this ADP (AMD Developer's Program) community.

Hackathon Submission

The final submission will happen on "Ultimatix Prime Events".

Following must be submitted –

1. **3–5 slide presentation** (PDF or PPT)

Slide 1 – Basic Information

- Team name
- Names of the team members / individual / role in the team for the Hackathon (Research, coding, building the business case, presentation, etc)
- Project Title
- Short Description

Slide 2 – Problem & Context

- Problem statement
- Target users / stakeholders
- Why this problem matters (business or operational relevance)
- Mapped hackathon challenge

Slide 3 – Solution Overview

- Solution architecture / workflow diagram
- AI approach used (e.g., GenAI, RAG, agents, vision, speech, multimodal)
- Key technologies / frameworks leveraged
- What was built during the hackathon

Slide 4

- Models used
- Dataset(s) used (if fine-tuning)
- Training time (if fine tuning)
- Number of tokens used for a couple of example scenarios
- End-to-end latency
- GPU usage including memory (gives a sense of right sizing)

Slide 5 – Impact & Demo Summary

- Expected impact or value (efficiency, productivity, scale, experience)
- Key differentiators / innovation
- Demo flow overview (what the jury should notice)

NOTE : The template is uploaded on the Prime Hub page

2. **Demo Recording**

Use one of the following recording tools which must be available on your device –

1. Microsoft Clipchamp
2. Google Vids

If there is no recording app present in your laptop/desktop, take the screenshots, save them as a pdf and submit

Understand ROCm (Radeon Open Compute) for the use-case development

ROCm is AMD’s open-source software platform for GPU computing, the AMD equivalent of NVIDIA's CUDA. It lets you run AI/ML workloads and HPC applications on AMD GPUs.

What developers commonly use it for:

- Running PyTorch and TensorFlow on AMD GPUs
- Porting CUDA-based workloads to AMD hardware
- Executing high-performance AI and ML workloads

Documentation

- ROCm documentation
- ROCm installation guide

What developers often use it for:

- Learning how to build and run AI workloads on AMD hardware
- Exploring fine-tuning and reinforcement learning workflows

Access

- Free access to all courses
- Includes hands-on labs with real GPU access
- Developed in partnership with DeepLearning.AI for advanced content

NOTE: Doubts will be clarified in the kick-off session planned on 8th June.